



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

Year 12

Trial Examination

2021

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Write your name on the front of every booklet.
- In Questions 11 to 14 show relevant mathematical reasoning and/or calculations.
- NESA approved calculators may be used.
- Weighting:30%

Section I Multiple Choice

- 10 marks
- Attempt all questions.
- Answer Sheet provided
- Allow about 15 minutes for this section

Section II Free Response

- 60 marks
- Start a separate booklet for each question.
- Each question is of equal value.
- All necessary working should be shown in every question.
- Allow about 1 hour and 45 minutes for this section.

Section I

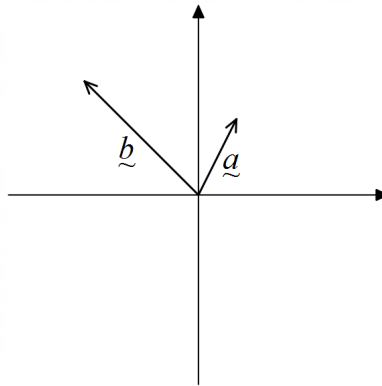
10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

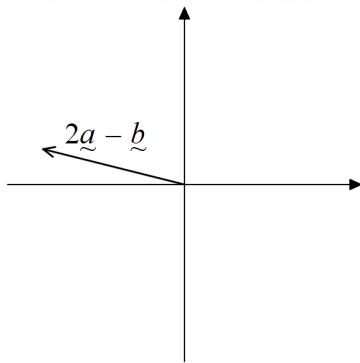
Use the multiple choice answer sheet for Questions 1 – 10.

Q1. Vectors $\underline{\underline{a}}$ and $\underline{\underline{b}}$ are shown in the diagram below.

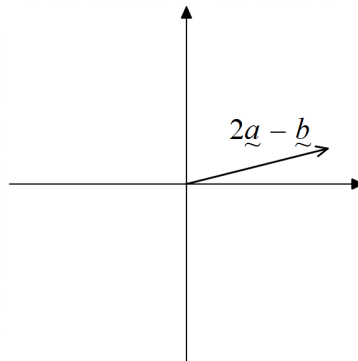


Which diagram best represents the vector $2\underline{\underline{a}} - \underline{\underline{b}}$?

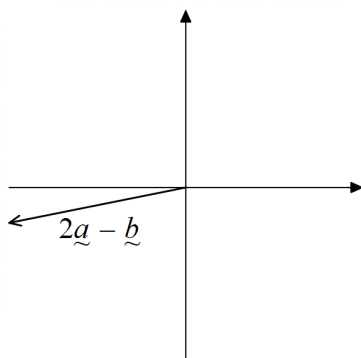
A



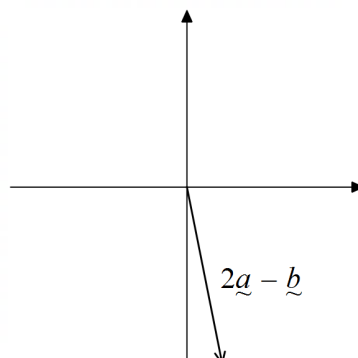
B



C



D



Q2. The radius of a circle is increasing at a constant rate of 0.4 meters per second. What is the rate of increase in the area of the circle at the instant when the circumference is 60π metres?

A. $0.24 \frac{m^2}{sec}$

B. $24 \frac{m^2}{sec}$

C. $24\pi \frac{m^2}{sec}$

D. $240\pi \frac{m^2}{sec}$

Q3. Which of the following expressions is equivalent to $3\cos x - \sqrt{3}\sin x$?

A. $2\sqrt{3}\cos\left(x - \frac{\pi}{3}\right)$

B. $2\sqrt{3}\cos\left(x + \frac{\pi}{3}\right)$

C. $2\sqrt{3}\cos\left(x - \frac{\pi}{6}\right)$

D. $2\sqrt{3}\cos\left(x + \frac{\pi}{6}\right)$

Q4. Find the equation to the tangent to the curve given by $x = 2y$ and $y = t^2 + 5$ at the point where $t = 2$

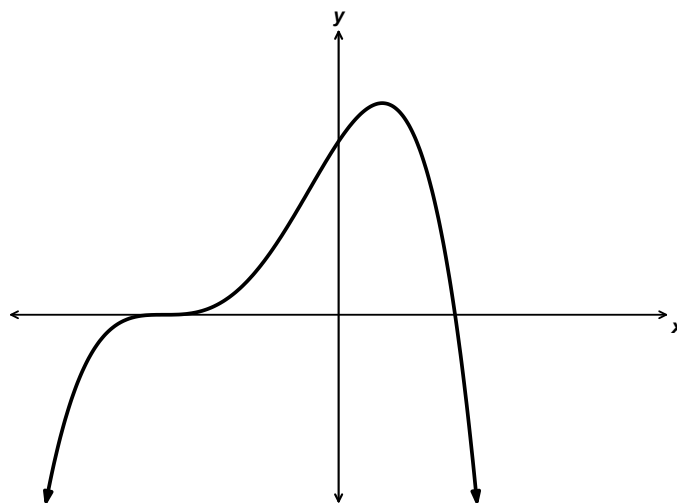
A. $y = 2x + 1$

B. $y = tx - 2t + 6$

C. $y = x + 4$

D. $y = x - 4$

Q5. The graph of a polynomial $y = P(x)$ is shown below.



What is the minimum possible degree of the polynomial $[P(x)]^2$?

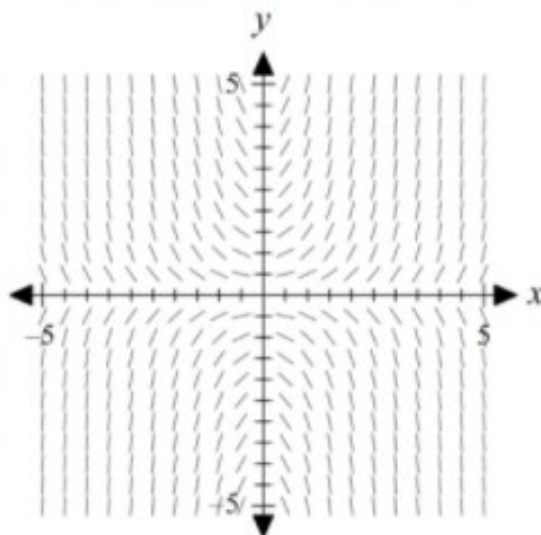
- A. 2
- B. 4
- C. 8
- D. 16

Q6. The variance of a Bernoulli distribution is $\frac{3}{16}$.

What is a possible value for the mean of the distribution?

- A. $\frac{5}{4}$
- B. $\frac{3}{4}$
- C. $\frac{3}{16}$
- D. $\frac{5}{16}$

Q7. The direction field of a differential equation is shown.



Which differential equation does the direction field best represent?

- A. $y' = \frac{x}{y}$
- B. $y' = xy$
- C. $y' = x^2 y^2$
- D. $y' = x + y$

Q8. Which of the following shows the correct substitution of the variable u into the integral

$$\int x\sqrt{3x-2} \, dx \text{ using } u = 3x-2?$$

- A. $\int x \left(u^{\frac{1}{2}} \right) dx$
- B. $\int \left(\frac{u+2}{3} \right) u^{\frac{1}{2}} du$
- C. $\frac{1}{9} \int (u+2) u^{\frac{1}{2}} du$
- D. $\frac{1}{3} \int \left(\frac{u}{3} + 2 \right) u^{\frac{1}{2}} du$

Q9. The region bounded by the y-axis and the curve $y = 6x - x^2 : D[0,3]$ is rotated around the y-axis. The volume of the resulting solid of revolution is given by:

- A. $\int_0^3 6x - x^2 \, dx$
- B. $\int_0^3 \pi(6x - x^2)^2 \, dx$
- C. $\int_0^9 \pi(3 + \sqrt{9 - y})^2 \, dy$
- D. $\int_0^9 \pi(3 - \sqrt{9 - y})^2 \, dy$

Q10. The scalar projection of the vector $\underline{\mathbf{u}} = \underline{\mathbf{i}} + 2\underline{\mathbf{j}}$ in the direction of the vector $\underline{\mathbf{v}}$ is 2.

Which of the following could be equal to $\underline{\mathbf{v}}$?

- A. $3\underline{\mathbf{i}} + 4\underline{\mathbf{j}}$
- B. $3\underline{\mathbf{i}} + 5\underline{\mathbf{j}}$
- C. $4\underline{\mathbf{i}} + 3\underline{\mathbf{j}}$
- D. $4\underline{\mathbf{i}} + 5\underline{\mathbf{j}}$

End of Multiple Choice

Section II

60 marks - Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

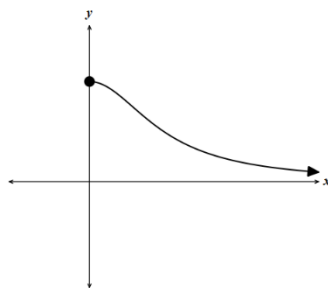
Question 11: Use new writing book for this question

15 marks

a) Solve for x given $\frac{x}{x-3} \geq 4$ **2**

b) Given $\cos x = -\frac{1}{3}$ and $\sin x > 0$ find the exact value of $\sin 2x$ **2**

c) The diagram shows the function $f(x) = \frac{1}{x^2 + 1}$ for $x \geq 0$.



State the domain and range of the inverse function $f^{-1}(x)$. **2**

d) Solve for x given $x^3 - 5x^2 + 8x - 4 = 0$ **3**

e) The ten members of a sporting team have the numbers 1 to 10 on the back of their jumpers.

i. Explain why, if six players are randomly chosen there must be at least one pair of players whose numbers add to 11. **1**

ii. What is the minimum number of players that need to be chosen to ensure that at least one pair of players have numbers that add to 12? **2**

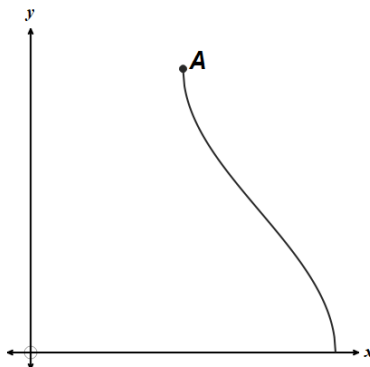
f) Use mathematical induction to prove **3**

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n} \quad \text{for integers } n \geq 1$$

End of Question 11

Question 12 : Use new writing book for this question**15 marks**

- a) The graph above below $y = \cos^{-1}(2x - 3)$. Determine the coordinates of point A . **2**



- b) In a dice game five dice are rolled simultaneously. One point is scored for each six showing face-up on the dice. Let X represent the score.

- (i) Explain why X can be considered as a Bernoulli variable **1**
- (ii) Find the expected value and variance of X **1**
- (iii) Find $P(X > 3)$, correct to four decimal places. **1**

- c) Find $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 2x \, dx$ in simplest exact form. **3**

- d) Using the substitution $x = 2\sin\theta$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, determine the exact value of

$$\int_0^1 \frac{1}{\sqrt{4-x^2}} \, dx \quad \mathbf{3}$$

- e) Find the solution of the differential equation $y' = 1 + 4y^2$ given $y(0) = \frac{1}{2}$ **4**

End of Question 12

Question 13 : Use new writing book for this question

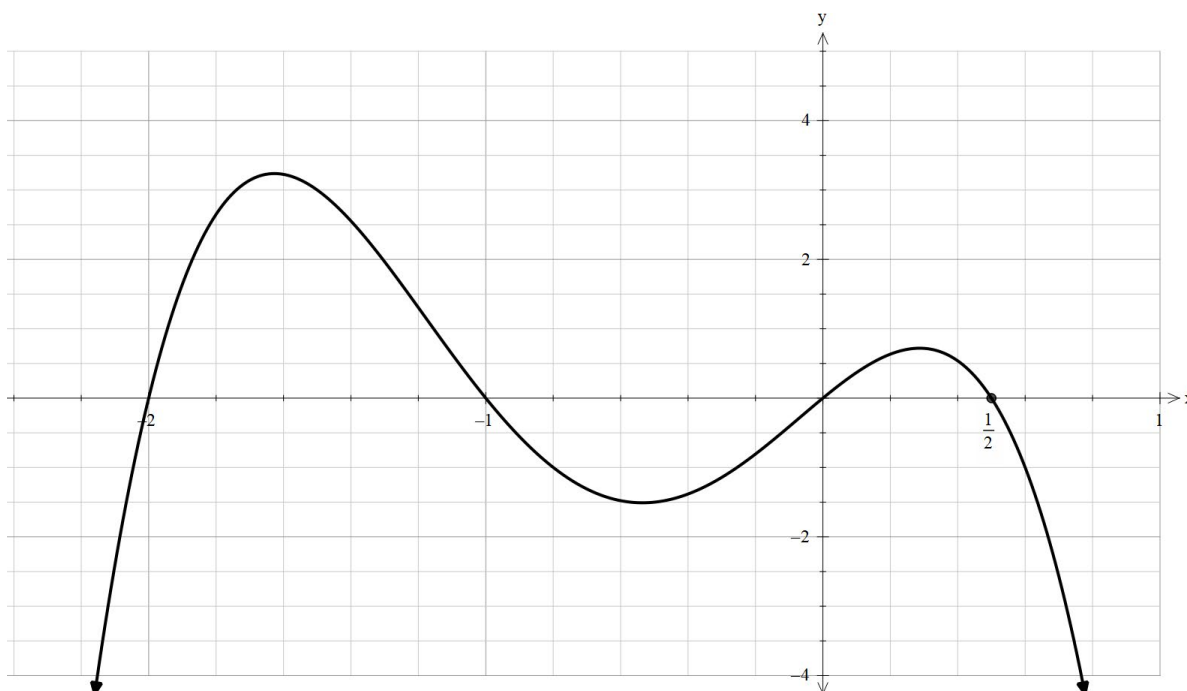
15 marks

- a) The curve $y = f(x)$ is shown below.
Using the Cartesian planes provided in the answer booklet, sketch the following graphs.

i. $y = f(|x|)$

ii. $y^2 = f(x)$

**1
3**



- b) Shares in a company X are particularly volatile. On any given trading day the probability that the share price will increase by \$1 is $\frac{3}{5}$ and the probability that the share price will fall by \$1 is $\frac{2}{5}$. If the share price is now \$6 find the probability that it will be less than \$6 after 5 trading days.

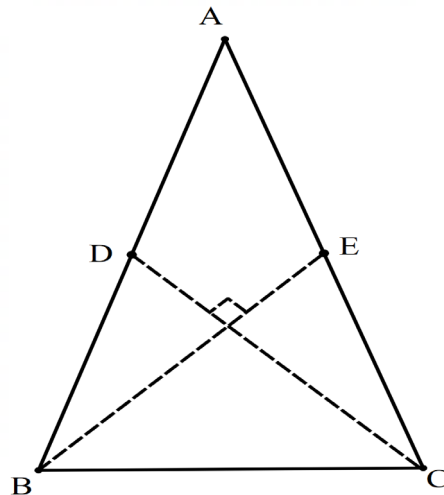
2

Question 13 continues on next page;

Question 13 continued.

- c) Determine the equation to the tangent to the curve $y = 3 \sin^{-1} \left(\frac{x}{2} \right)$ at $x = 1$ **3**

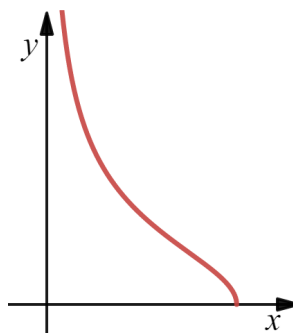
- d) In the isosceles triangle ABC , $|\overrightarrow{AB}| = |\overrightarrow{AC}|$. D is the midpoint of side AB and E is the midpoint of side AC . $\overrightarrow{CD}$ is perpendicular to $\overrightarrow{BE}$.



Use **vector methods** to find the size of $\angle BAC$.

3

- e) Part of the graph of the function $f(x) = \sqrt{\frac{1 - 3x^2}{4x^2}}$ is shown below.



The region bounded by the curve $y = f(x)$, $y = \frac{3}{2}$ and the coordinate axes is rotated about the y -axis. Find the exact volume of the solid formed.

3

End of Question 13.

a) i. Show $\sin(x - y)\sin(x + y) = \sin^2 x - \sin^2 y$. 2

ii. Solve $\sin^2 3x - \sin^2 x = \sin 4x$ for $0 \leq x \leq \pi$ 2

b) The population $P(t)$ of bacteria in a petri dish is modelled by the logistic differential equation

$$\frac{dP}{dt} = \frac{P}{6} \left(1 - \frac{P}{8000} \right) \text{ where } P(0) = P_0 \text{ and } t \text{ is the time in hours.}$$

i. If the initial population P_0 is 1000 bacteria, show that $P(t) = \frac{8000}{1 + 7e^{-\frac{t}{6}}}$ 3

(You may use the fact that $\frac{Q}{P(Q - P)} = \frac{1}{P} + \frac{1}{Q - P}$)

ii. If instead the initial population P_0 were 12000 bacteria, describe what would have happened to the population as t increases. 1

c) The vectors $\underline{\mathbf{p}} = \begin{pmatrix} a \\ 3 \end{pmatrix}$ and $\underline{\mathbf{q}} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ are parallel.

i. Find a . 1

ii. The vector $\underline{\mathbf{r}} = \begin{pmatrix} x \\ y \end{pmatrix}$ is perpendicular to vector $\underline{\mathbf{q}}$ and $|\underline{\mathbf{q}} - \underline{\mathbf{r}}| = \sqrt{65}$.

Find the possible values of x and y . 3

d) Using the expansion of $(1 + x)^{2n}$ and given that 3

$${}^{2n}\mathbf{C}_1 + 2({}^{2n}\mathbf{C}_2) + 3({}^{2n}\mathbf{C}_3) + \dots + 2n({}^{2n}\mathbf{C}_{2n}) = n \times 2^{2n}$$

Show that

$$2({}^{2n}\mathbf{C}_1) + 3({}^{2n}\mathbf{C}_2) + 4({}^{2n}\mathbf{C}_3) + \dots + (2n + 1)({}^{2n}\mathbf{C}_{2n}) = (n + 1) \times 2^{2n} - 1$$

End of Examination

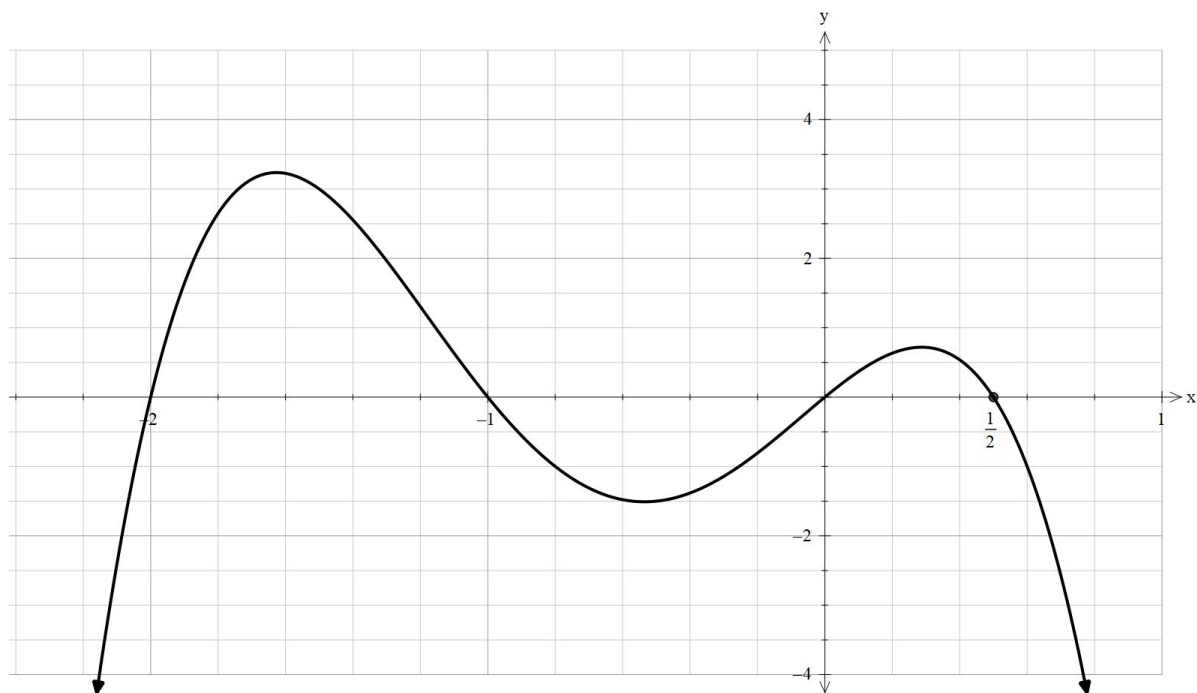
Student Number:

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Answer sheet for Question 13b.

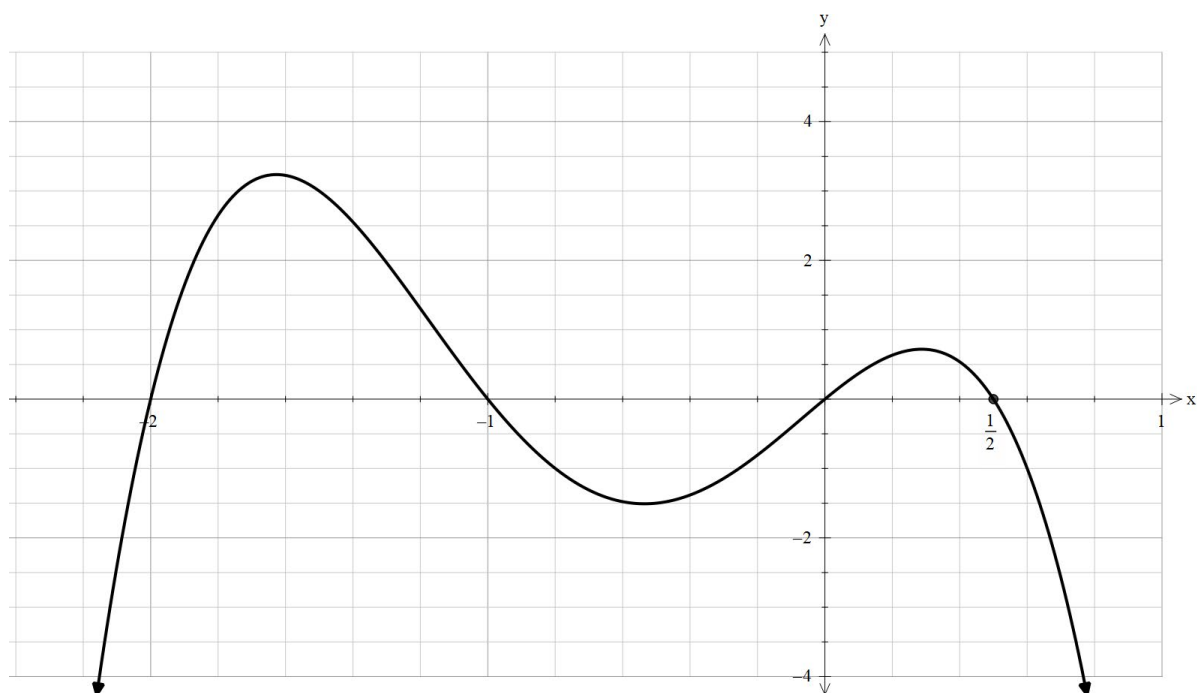
i. $y = f(|x|)$

1



ii. $y^2 = f(x)$

3





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Question 11: Use new writing book for this question

15 marks

a) Solve for x given $\frac{x}{x-3} \geq 4$

2

$$\frac{x}{x-3} \geq 4$$
$$x \neq 3$$

$$\frac{x}{x-3} = 4$$

$$x = 4x - 12$$

$$3x = 12$$

$$x = 4$$

Test regions

$$3 < x \leq 4$$

2 marks correct answer , one mark correct significant points

b) Given $\cos x = -\frac{1}{3}$ and $\sin x > 0$ find the exact value of $\sin 2x$

2

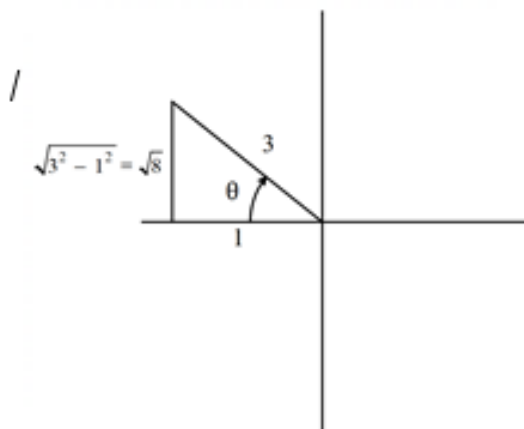
Solution

$$\sin 2x = 2 \cos x \sin x$$

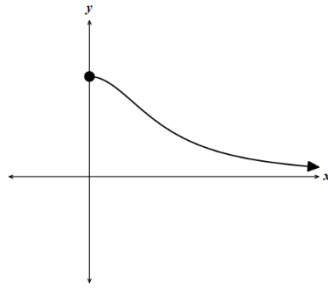
$$= 2 \times -\frac{1}{3} \times \frac{\sqrt{8}}{3}$$

$$= -\frac{2\sqrt{8}}{9}$$

$$= -\frac{4\sqrt{2}}{9}$$



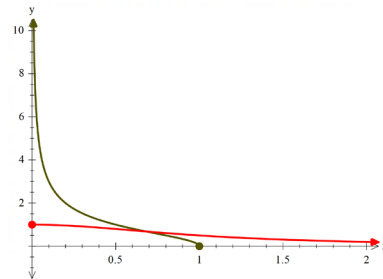
- c) The diagram shows the function $f(x) = \frac{1}{x^2 + 1}$ for $x \geq 0$.



State the domain and range of the inverse function $f^{-1}(x)$.

2

$$D : (0, 1] \quad R : [0, \infty)$$



- d) Solve for x given $x^3 - 5x^2 + 8x - 4 = 0$

3

$$P(1) = 1 - 5 + 8 - 4 = 0 \Rightarrow (x - 1) \text{ factor}$$

$$P(2) = 8 - 20 + 16 - 4 = 0 \Rightarrow (x - 2) \text{ factor}$$

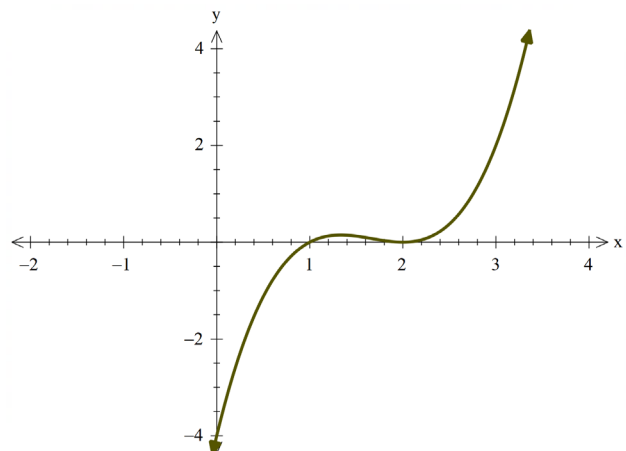
$$P(x) = (x - 1)(x - 2)(x \pm c)$$

$$\therefore -4 = -1 \times -2 \times c$$

$$c = -2$$

$$P(x) = (x - 1)(x - 2)^2$$

$$x = 1 \text{ or } 2$$



- e) The ten members of a sporting team have the numbers 1 to 10 on the back of their jumpers.
- i. Explain why, if six players are randomly chosen there must be at least one pair of players whose numbers add to 11. 1
 - ii. What is the minimum number of players that need to be chosen to ensure that at least one pair of players have numbers that add to 12? 2

Pairs that make 11

(1,10) (2,9) (3,8) (4,7) (5,6) - if the first 5 numbers are picked one from each pair to avoid the sum of 11, the 6th number must complete a pair adding to 11.

(2,10) (3,9) (4,8) (1,11) (5,7) with 1 and 6 remaining

To avoid a pair – max number would be 1, 6 plus one each from the pairs = $2+5=7$

Therefore one more selection must produce a sum of 12. = 8

f) Use mathematical induction to prove

3

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n} \quad \text{for integers } n \geq 1$$

Sample answer:

Step 1 Prove true for $n=1$

$$LHS = \frac{1}{2}$$

$$RHS = 2 - \frac{1+2}{2} = \frac{1}{2}$$

$\therefore$ true for $n=1$

Step 2 Let $n=k$ be a value for which the statement is true
ie

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k}$$

Step 3 Prove true for $n=k+1$

i.e prove

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}} = 2 - \frac{(k+1)+2}{2^{k+1}} = 2 - \frac{k+3}{2^{k+1}}$$

$$\begin{aligned} LHS &= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} \\ &= 2 - \left(\frac{2(k+2)}{2^{k+1}} - \frac{k+1}{2^{k+1}} \right) \\ &= 2 - \left(\frac{2k+4-k-1}{2^{k+1}} \right) \\ &= 2 - \frac{k+3}{2^{k+1}} \text{ as required.} \end{aligned}$$

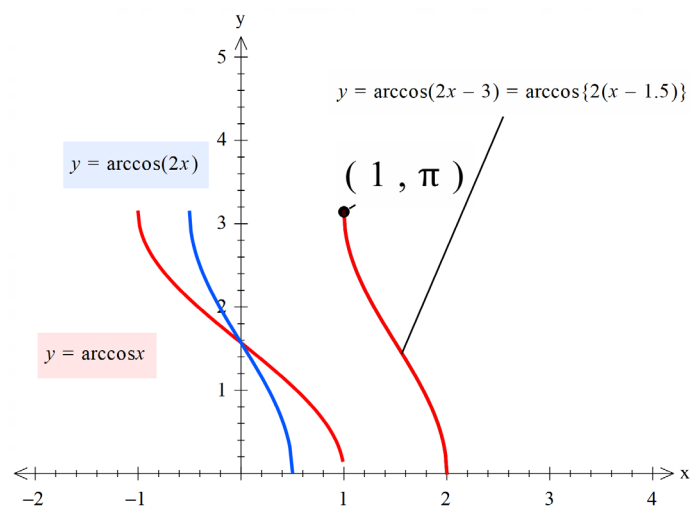
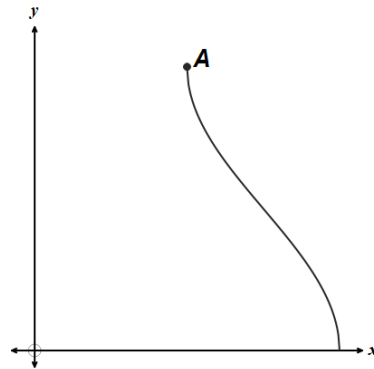
$\therefore$ true by mathematical induction

Question 12 : Use new writing book for this question

15 marks

- a) The graph above below $y = \cos^{-1}(2x - 3)$. Determine the coordinates of point A .

2



- b) In a dice game five dice are rolled simultaneously. One point is scored for each six showing face-up on the dice. Let X represent the score.

- | | | |
|-------|---|---|
| (i) | Explain why X can be considered as a Bernoulli variable | 1 |
| (ii) | Find the expected value and variance of X | 1 |
| (iii) | Find $P(X > 3)$, correct to four decimal places. | 1 |

X has a binomial distribution. Its shape is positively skewed.
$E(X) = np = 5 \times \frac{1}{6} = \frac{5}{6}$ (see Reference Sheet) $\text{Var}(X) = np(1 - p) = 5 \times \frac{1}{6} \times \frac{5}{6} = \frac{25}{36}$ (see Reference Sheet)
$P(X > 3) = P(X = 4 \text{ or } X = 5) = P(X = 4) + P(X = 5)$ $= \binom{5}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1 + \binom{5}{5} \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^0$ or $\binom{5}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1 + \left(\frac{1}{6}\right)^5$ (see Reference Sheet) $= 0.0033$ (4 d.p.)

- c) Find $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 2x \, dx$ in simplest exact form. 3

$$\begin{aligned}
 & \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 2x \, dx \\
 &= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (1 - \cos 4x) \, dx \\
 &= \frac{1}{2} \left[x - \frac{1}{4} \sin 4x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\
 &= \frac{1}{2} \left[\left(\frac{\pi}{3} - \frac{1}{4} \sin \frac{4\pi}{3} \right) - \left(\frac{\pi}{4} - \frac{1}{4} \sin \pi \right) \right] \\
 &= \frac{1}{2} \left(\frac{\pi}{6} + \frac{1}{4} \times \frac{\sqrt{3}}{2} \right)
 \end{aligned}$$

- d) Using the substitution $x = 2\sin\theta$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, determine the exact value of

$$\int_0^1 \frac{1}{\sqrt{4-x^2}} dx$$

3

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{4-x^2}} dx$$

$$x = 2\sin\theta$$

$$x^2 = 4\sin^2\theta$$

$$x = 0 \Rightarrow 0 = 2\sin\theta \Rightarrow \theta = 0$$

$$x = 1 \Rightarrow \frac{1}{2} = \sin\theta \Rightarrow \theta = \frac{\pi}{6}$$

$$dx = 2\cos\theta d\theta$$

$$\int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{4-4\sin^2\theta}} \times 2\cos\theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{2} \frac{1}{\sqrt{1-\sin^2\theta}} 2\cos\theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{2} \times \frac{1}{\cos\theta} \times 2\cos\theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} 1 d\theta$$

$$= [\theta]_0^{\frac{\pi}{6}}$$

$$= \frac{\pi}{6}$$

e) Find the solution of the differential equation given $y(0) = \frac{1}{2}$

4

$$y' = 1 + 4y^2$$

$$\frac{dy}{dx} = 1 + 4y^2$$

$$\int \frac{1}{1 + 4y^2} dy = \int dx$$

$$\tan^{-1}(2y) = x + c$$

$$x = 0 \Rightarrow y = \frac{1}{2}$$

$$\tan^{-1}\left(2 \times \frac{1}{2}\right) = 0 + c$$

$$c = \frac{\pi}{4}$$

$$2y = \tan\left(x + \frac{\pi}{4}\right)$$

$$y = \frac{1}{2}\tan\left(x + \frac{\pi}{4}\right)$$

End of Question 12

a) The curve $y = f(x)$ is shown below.

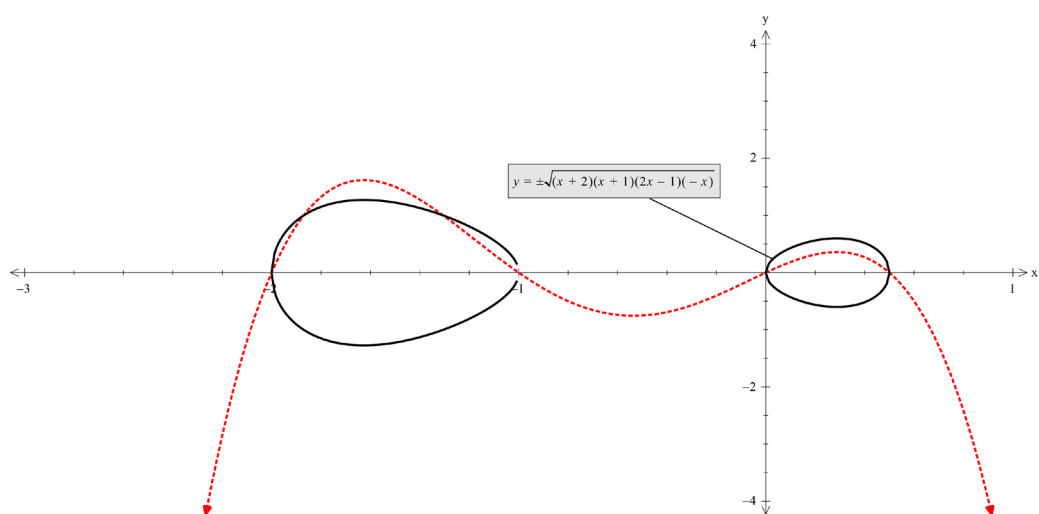
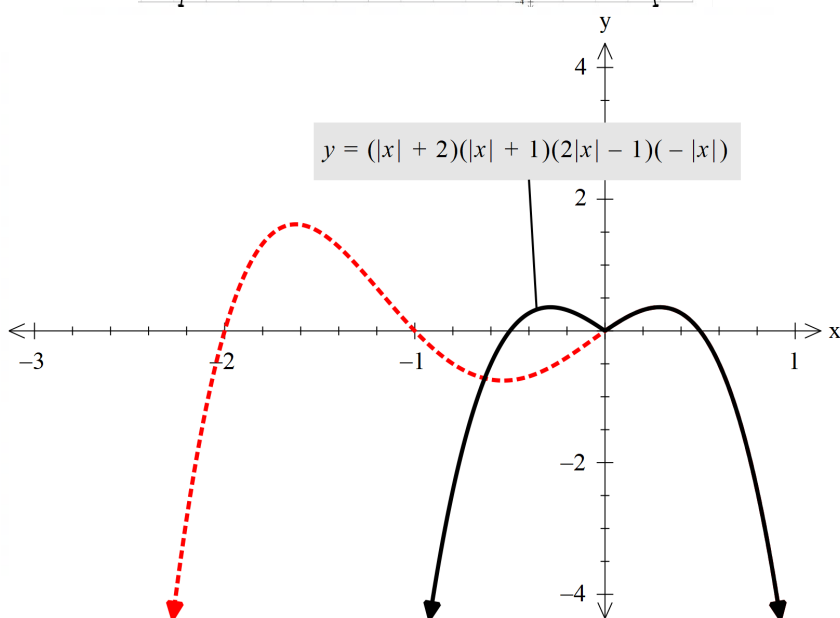
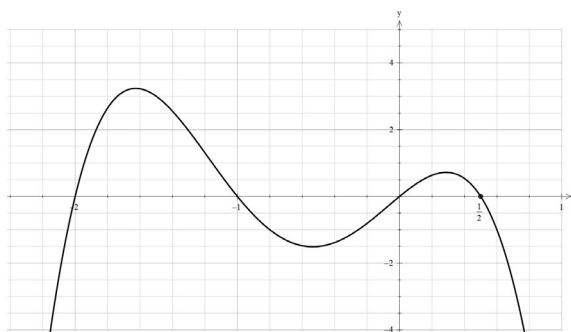
Using the Cartesian planes provided in the answer booklet, sketch the following graphs.

i. $y = f(|x|)$

1

ii. $y^2 = f(x)$

3



- b) Shares in a company X are particularly volatile. On any given trading day the probability that the share price will increase by \$1 is $\frac{3}{5}$ and the probability that the share price will fall by \$1 is $\frac{2}{5}$. If the share price is now \$6 find the probability that it will be less than \$6 after 5 trading days.

2

Criteria	Marks
• Provides correct answer.	2
• Some attempt to write in binomial probability form or equivalent progress.	1

Sample answer:

$$\begin{aligned}
 \text{Prob}(< \$6 \text{ after 5 days}) &= \text{Prob}(\text{Majority of movements are down}) \\
 &= \text{Prob}(\text{number of up movements} \leq 2) \\
 &= \left(\frac{2}{5}\right)^5 + {}^5C_1 \left(\frac{2}{5}\right)^4 \left(\frac{3}{5}\right)^1 + {}^5C_2 \left(\frac{2}{5}\right)^3 \left(\frac{3}{5}\right)^2 \\
 &= \frac{992}{3125}
 \end{aligned}$$

$$\text{Let } p(\text{Increase}) = \frac{3}{5}$$

$$\text{Let } q(\text{Decrease}) = \frac{2}{5}$$

Let n = no of trials

$$(p + q)^n$$

To be less than 6 dollars after 5 trading days must decrease on 3 or more days.

ie

Case 1 $\Rightarrow$ IIIII = \$5 increase

Case 2 $\Rightarrow$ II IID = \$3 increase

Case 3 $\Rightarrow$ II I DD = \$1 increase

Case 4 $\Rightarrow$ II D DD = \$1 Decrease

Case 5 $\Rightarrow$ I D D D D = \$3 decrease

Case 6 $\Rightarrow$ D D D D D = \$5 Decrease

$$(p + q)^5 = {}^5C_0 \left(\frac{3}{5}\right)^5 + {}^5C_1 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right) + {}^5C_2 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2 + {}^5C_3 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^3 + {}^5C_4 \left(\frac{3}{5}\right) \left(\frac{2}{5}\right)^4 + {}^5C_5 \left(\frac{2}{5}\right)^5$$

$$P(< \$6) = {}^5C_3 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^3 + {}^5C_4 \left(\frac{3}{5}\right) \left(\frac{2}{5}\right)^4 + {}^5C_5 \left(\frac{2}{5}\right)^5$$

Question 13 continued.

- c) Determine the equation to the tangent to the curve $y = 3 \sin^{-1} \left(\frac{x}{2} \right)$ at $x = 1$

3

Determine the equation to the tangent to the curve $y = 3 \sin^{-1} \left(\frac{x}{2} \right)$ at $x = 1$

$$y = 3 \sin^{-1} \left(\frac{x}{2} \right) \text{ at } x = 1$$

at $x = 1$

$$y = 3 \sin^{-1} \frac{1}{2} = 3 \times \frac{\pi}{6} = \frac{\pi}{2}$$

$$\frac{d \left(\sin^{-1} \frac{f(x)}{a} \right)}{dx} = \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}$$

$$\therefore \frac{dy}{dx} = \frac{3}{\sqrt{4 - x^2}}$$

$$\text{at } x = 1 \quad m = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$y - y_1 = \sqrt{3} (x - x_1)$$

$$y - \frac{\pi}{2} = \sqrt{3} (x - 1)$$

$$y = \sqrt{3} x + \left(\frac{\pi}{2} - \sqrt{3} \right)$$

3 marks – correct solution

2 marks

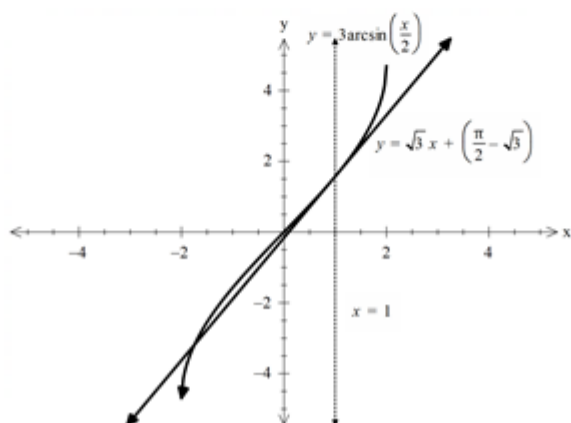
- Correct gradient from correct derivative but no further progress.
- Correct equation from a gradient correctly formed from incorrect derivative.

1 mark

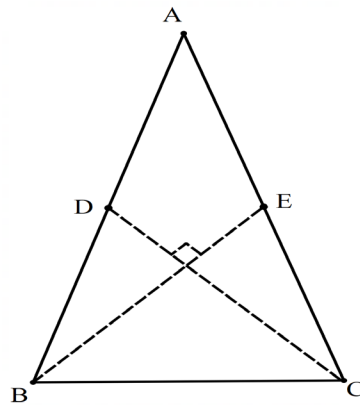
Any of

Correct y

Correct derivative



- d) In the isosceles triangle ABC , $|\overrightarrow{AB}| = |\overrightarrow{AC}|$. D is the midpoint of side AB and E is the midpoint of side AC . $\overrightarrow{CD}$ is perpendicular to $\overrightarrow{BE}$.



Use **vector methods** to find the size of $\angle BAC$.

3

Let

$$\begin{aligned}\overrightarrow{AB} &= \underline{\underline{a}} \\ \overrightarrow{AC} &= \underline{\underline{b}}\end{aligned}$$

$$|\underline{\underline{a}}| = |\underline{\underline{b}}| = x$$

$$\overrightarrow{ED} = \underline{\underline{a}} - \frac{1}{2}\underline{\underline{b}}$$

$$\overrightarrow{DC} = \underline{\underline{b}} - \frac{1}{2}\underline{\underline{a}}$$

$$\overrightarrow{ED} \cdot \overrightarrow{DC} = 0 \text{ as vectors perpendicular}$$

$$\left(\underline{\underline{a}} - \frac{1}{2}\underline{\underline{b}}\right) \cdot \left(\underline{\underline{b}} - \frac{1}{2}\underline{\underline{a}}\right) = \underline{\underline{a}} \cdot \underline{\underline{b}} - \frac{1}{2}\underline{\underline{a}} \cdot \underline{\underline{a}} - \underline{\underline{b}} \cdot \frac{1}{2}\underline{\underline{b}} + \frac{1}{4}\underline{\underline{a}} \cdot \underline{\underline{b}}$$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta = |\underline{a}|^2 \cos \theta = x^2 \cos \theta$$

$$\underline{a} \cdot \underline{a} = |\underline{a}|^2 = x^2$$

$$\underline{b} \cdot \underline{b} = |\underline{b}|^2 = x^2$$

$$\frac{5}{4} \underline{a} \cdot \underline{b} - \frac{1}{2} \underline{a} \cdot \underline{a} - \frac{1}{2} \underline{b} \cdot \underline{b} = 0$$

$$\frac{5}{4} x^2 \cos \theta - x^2 = 0$$

$$\frac{5}{4} \cos \theta - 1 = 0$$

$$\cos \theta = \frac{4}{5}$$

$$\theta = \cos^{-1} \frac{4}{5} = 36.87^\circ$$

3 marks – full solution

2 marks – using dot product to achieve this equation or similar

$$\frac{5}{4} \underline{a} \cdot \underline{b} - \frac{1}{2} \underline{a} \cdot \underline{a} - \frac{1}{2} \underline{b} \cdot \underline{b} = 0$$

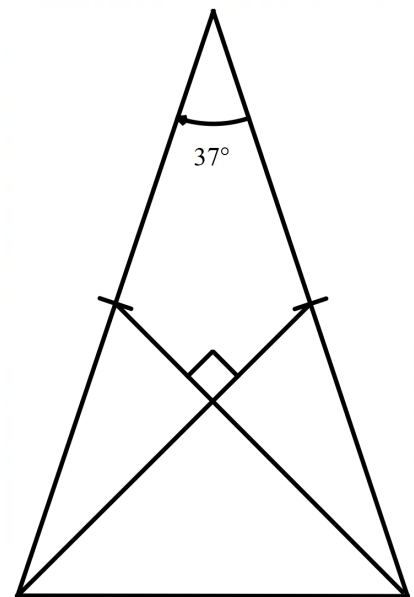
1 mark

one of

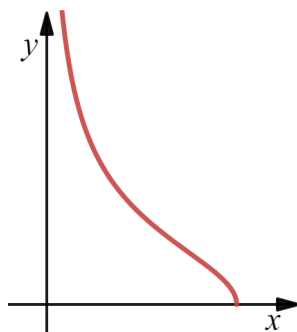
- Recognition $\underline{ED} \cdot \underline{DC} = 0$ as perpendicular
- Show $\underline{ED} = \underline{a} - \frac{1}{2} \underline{b}$
- Show $\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta = |\underline{a}|^2 \cos \theta$

0 marks

Non-vector solution



iPart of the graph of the function $f(x) = \sqrt{\frac{1-3x^2}{4x^2}}$ is shown below.



The region bounded by the curve $y = f(x)$, $y = \frac{3}{2}$ and the coordinate axes is rotated about the y -axis. Find the exact volume of the solid formed.

Criteria	Marks
• Provides correct solution	3
• Finds the inverse-trigonometric primitive of the integral, or equivalent merit	2
• Determines integral that yields volume, or equivalent merit	1

Sample answer:

$$y = \sqrt{\frac{1-3x^2}{4x^2}}$$

$$y^2 = \frac{1-3x^2}{4x^2}$$

$$4x^2y^2 = 1-3x^2$$

$$x^2(4y^2+3) = 1$$

$$x^2 = \frac{1}{4y^2+3}$$

$$\text{Now, } V = \pi \int_0^{\frac{3}{2}} x^2 dy$$

$$\therefore V = \pi \int_0^{\frac{3}{2}} \frac{1}{4y^2+3} dy$$

$$= \frac{\pi}{2\sqrt{3}} \left[\tan^{-1} \frac{2y}{\sqrt{3}} \right]_0^{\frac{3}{2}}$$

$$= \frac{\pi}{2\sqrt{3}} [\tan^{-1} \sqrt{3} - \tan^{-1} 0]$$

$$= \frac{\pi^2}{6\sqrt{3}} \text{ units}^3$$

End of Question 13.

Question 14 : Use new writing book for this question

15 Marks

- a) i. Show $\sin(x-y)\sin(x+y) = \sin^2 x - \sin^2 y$.

2

Sample answer:

$$\begin{aligned}\sin(x-y)\sin(x+y) &= \frac{1}{2}[\cos[(x-y)-(x+y)] - \cos[(x-y)+(x+y)]] \\ &= \frac{1}{2}[\cos(-2y) - \cos(2x)] \\ &= \frac{1}{2}[\cos(2y) - \cos(2x)] \\ &= \frac{1}{2}[(1 - 2\sin^2 y) - (1 - 2\sin^2 x)] \\ &= \sin^2 x - \sin^2 y\end{aligned}$$

- ii. Solve $\sin^2 3x - \sin^2 x = \sin 4x$ for $0 \leq x \leq \pi$

2

Sample answer:

$$\begin{aligned}\text{From (i) } \sin^2 3x - \sin^2 x &= \sin(3x-x)\sin(3x+x) = \sin 2x \sin 4x \\ \sin 2x \sin 4x &= \sin 4x \text{ for } 0 \leq x \leq \pi \\ \sin 2x \sin 4x - \sin 4x &= 0 \\ \sin 4x(\sin 2x - 1) &= 0 \\ \sin 4x = 0 &\Rightarrow 4x = 0, \pi, 2\pi, 3\pi, 4\pi \\ x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi \\ \sin 2x = 1 &\Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4} \\ \therefore x &= 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi\end{aligned}$$

- b) The population $P(t)$ of bacteria in a petri dish is modelled by the logistic differential equation

$$\frac{dP}{dt} = \frac{P}{6} \left(1 - \frac{P}{8000} \right) \text{ where } P(0) = P_0 \text{ and } t \text{ is the time in hours.}$$

- i. If the initial population P_0 is 1000 bacteria, show that $P(t) = \frac{8000}{1 + 7e^{-\frac{t}{6}}}$

3

$$\frac{dP}{dt} = \frac{P}{6} \left(1 - \frac{P}{8000} \right)$$

$$\frac{dt}{dP} = \frac{48000}{P(8000 - P)}$$

$$\int dt = 6 \int \frac{8000}{P(8000 - P)} dP$$

$$\frac{1}{6} \int dt = \int \left(\frac{1}{P} + \frac{1}{8000 - P} \right) dP \text{ using } \frac{Q}{P(Q - P)} = \frac{1}{P} + \frac{1}{Q - P}$$

$$\frac{1}{6}t + c = \ln|P| - \ln|8000 - P|$$

$$\frac{t}{6} + c = \ln \left| \frac{P}{8000 - P} \right|$$

$$\left| \frac{P}{8000 - P} \right| = e^c e^{\frac{t}{6}}$$

$$\frac{P}{8000 - P} = \pm e^c e^{\frac{t}{6}}$$

$$\frac{P}{8000 - P} = A e^{\frac{t}{6}}, \text{ where } A = \pm e^c$$

When $t = 0$, $P = 1000$:

$$\frac{1000}{8000 - 1000} = A e^0$$

$$\therefore A = \frac{1}{7}$$

$$\therefore \frac{P}{8000 - P} = \frac{1}{7} e^{\frac{t}{6}}$$

$$\therefore P = \frac{8000}{1 + 7e^{-\frac{t}{6}}}$$

(You may use the fact that $\frac{Q}{P(Q - P)} = \frac{1}{P} + \frac{1}{Q - P}$)

- ii. If instead the initial population P_0 were 12000 bacteria, describe what would have happened to the population as t increases.

1

Sample answer:

The carrying capacity for the population is 8000. If $P(0) > 8000$, such as 12000, the population will decay and approach the asymptotic population size of 8000 as time increases.

Alternatively: The equation for $\frac{dP}{dt}$ shows that when $P = 12000$, $\frac{dP}{dt} < 0$ (population will decrease). As P decreases and approaches 8000, $\frac{dP}{dt}$ approaches zero. Thus, the population will decrease and will approach 8000.

c) The vectors $\underline{\mathbf{p}} = \begin{pmatrix} a \\ 3 \end{pmatrix}$ and $\underline{\mathbf{q}} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ are parallel.

i. Find a .

1

$$\text{Parallel} \Rightarrow \lambda \begin{pmatrix} a \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$3\lambda = 4 \Rightarrow \lambda = \frac{4}{3}$$

$$\frac{4}{3}a = 2$$

$$a = \frac{3}{2}$$

ii. The vector $\underline{\mathbf{r}} = \begin{pmatrix} x \\ y \end{pmatrix}$ is perpendicular to vector $\underline{\mathbf{q}}$ and $|\underline{\mathbf{q}} - \underline{\mathbf{r}}| = \sqrt{65}$.

Find the possible values of x and y .

3

$$\text{Perpendicular} \Rightarrow \lambda \begin{pmatrix} 2 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$2x + 4y = 0$$

$$x = -2y$$

$$\underline{\mathbf{q}} - \underline{\mathbf{r}} = \begin{pmatrix} 2 - x \\ 4 - y \end{pmatrix} = \begin{pmatrix} 2 + 2y \\ 4 - y \end{pmatrix}$$

$$|\underline{\mathbf{q}} - \underline{\mathbf{r}}| = \sqrt{65} \Rightarrow \sqrt{(2 + 2y)^2 + (4 - y)^2} = \sqrt{65}$$

squaring

$$4 + 8y + 4y^2 + 16 - 8y + y^2 = 65$$

$$5y^2 + 20 = 65$$

$$y^2 = 9$$

$$y = \pm 3$$

$$\therefore x = 6, y = -3 \text{ or } x = -6, y = 3$$

d) Using the expansion of $(1+x)^{2n}$ and given that

3

$$2^n C_1 + 2(2^n C_2) + 3(2^n C_3) + \dots + 2n(2^n C_{2n}) = n \times 2^{2n}$$

Show that

$$2(2^n C_1) + 3(2^n C_2) + 4(2^n C_3) + \dots + (2n+1)(2^n C_{2n}) = (n+1) \times 2^{2n} - 1$$

Sample answer:

$$(1+x)^{2n} = \binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n}$$

substitute $x=1$

$$\binom{2n}{0} + \binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n} = 2^{2n}$$

$$\binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n} = 2^{2n} - \binom{2n}{0}$$

$$\binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n} = 2^{2n} - 1$$

$$2\binom{2n}{1} + 3\binom{2n}{2} + \dots + (2n+1)\binom{2n}{2n} = \left[\binom{2n}{1} + 2\binom{2n}{2} + \dots + 2n\binom{2n}{2n} \right] + \left[\binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n} \right]$$

$$= n \times 2^{2n} + 2^{2n} - 1$$

$$= (n+1) \times 2^{2n} - 1$$

End of Examination

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